Fiscal Year:	FY 2016	**Task Last Updated:**	FY 03/10/2016
PI Name:	Gur, Ruben C. Ph.D.		
Project Title:	Cognition Validation Study In High-Performing Healthy Controls at Penn		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Neurobehavioral and Psychosocial Factors Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Address 1:	Hospital of the University of Pennsylvania		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-4283	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	11/01/2014	**End Date:**	10/31/2015
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Basner, Mathias M.D., Ph.D. (University of Pennsylvania)		
Grant/Contract No.:	NCC 9-58-NBPF00009		
Performance Goal No.:			
Performance Goal Text:			

This project addresses the need of the National Space Biomedical Research Institute (NSBRI) Neurobehavioral and Psychosocial Factors (NBPF) Team to validate the entirety of the Cognition battery against the currently used WinSCAT, as well as the need to validate the iPad version of the battery, on a sample of high-performing individuals comparable to that of astronauts. While the battery consists of well-validated tests, the battery itself has never been validated in its current implementation. This study will allow for validation in reliability, sensitivity to age and gender, and correlations among the individual tests. High-performing subjects are needed for the validation of the battery for comparability to the space and space analog travelers for whom the Cognition battery is intended. Pulsar has created an iPad version of the Cognition battery, which allows for greater portability and would be advantageous in multiple foreseeable conditions. This study is needed to compare performance on a laptop to that on an iPad. The touchscreen platform should produce consistently faster response times, but this constant needs to be evaluated on the different tests. Cognition contains 10 individual tests, which will be administered twice (two test versions) to 96 participants, once on

Task Description:	the iPad and once on the laptop (counterbalanced across subjects). The battery tests several domains including sensory-motor speed, memory, visual abstraction, spatial orientation, emotion recognition, attention, and risk-related decision-making. Reliability and validation were assessed using Classical Test Theory and Item Response Theory. We determined correlations among an individual test's items and that each test has the appropriate psychometric structure. Participants were also administered the WinSCAT on the laptop computer. Since initiating the project at the beginning of this year, all 96 individuals (48 males/48 females) have been enrolled and tested with the average female age being 40.35 and the average male age being 40.27. The stipulated goal of enrolling 96 individuals in four age bins (25-32, 33-40, 41-48, and 49-59) balanced by sex was precisely met. The completion rate for all three neurocognitive batteries was 100%. In all, 288 tests were successfully administered. Despite initial delays, this project was completed on time and all its recruitment goals were met. We demonstrated the feasibility of recruiting within one year a cohort of high performing individuals, with Masters degrees and higher, across the age range and sex distribution expected from space and space analog travelers. The results have supported the sensitivity of Cognition to age effects and sex differences even within this high-performing sample and the narrow age range. We can translate between versions. Further validation of Cognition can be achieved by repeated measures, and this sample has consented to be re-contacted. Such a study will be extremely relevant to the current studies of Cognition, in which it is administered repeatedly to space and space analog travelers. The present cohort could also be invaluable in establishing data on brain-behavior relations. Multiple current studies using Cognition apply it in conjunction with structural and functional neuroimaging. Penn has first-rate facilities for neuroimaging, and the current sample will be easily recruited for neuroimaging studies. We could establish an invaluable resource for gauging effects of space and space analog travel against a reliable normative base. The functional neuroimaging studies could include a version of Cognition, in which information on regional brain activation can be obtained concurrently with measures of performance (see Roalf et al., 2014). This will undoubtedly be the methodology of the future, and with NSBRI help we can make it happen now. Reference Roalf DR, Ruparel K, Gur RE, Bilker W, Gerraty R, Elliott MA, Gallagher RS, Almasy L, Pogue-Geile MF, Prasad K, Wood J, Nimgaonkar VL, Gur RC. Neuroimaging predictors of cognitive performance across a standardized neurocognitive battery. Neuropsychology. 2014, 28(2), 161-76.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will provide a dynamic and unobtrusive test environment able to assess cognitive performance of high-performing adults. Cognition was developed to be used in space by astronauts; however, there are Earth-based benefits that could be provided by the battery. The cognition battery, once validated through this project, would provide assessments of cognitive functioning in high-performing adults who carry out high stress jobs. It would offer milestone performance scores in areas such as decision making, mental abstraction, sensory-motor skills, and working memory that would be necessary in these fields.
Task Progress:	During the last funded year, the Cognition Validation Project was completed. The expected 96 individuals in their respective age bins were enrolled and all completed all testing.
Bibliography Type:	Description: (Last Updated: 03/13/2018)
Articles in Peer-reviewed Journals	Basner M, Savitt A, Moore TM, Port AM, McGuire S, Ecker AJ, Nasrini J, Mollicone DJ, Mott CM, McCann T, Dinges DF, Gur RC. "Development and validation of the Cognition test battery for spaceflight." Aerospace Medicine and Human Performance. 2015 Nov;86(11):942-52. http://dx.doi.org/10.3357/AMHP.4343.2015 ; PubMed PMID: 26564759; PubMed Central PMCID: PMC4691281 , Nov-2015